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Methodological Guidelines for Design Science Research

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Abstract

Design science research is typically used to conduct research in the information systems space. Design science research has also existed for many years, and might be seen as a daunting task when first undertaken by inexperienced researchers. This is most likely due to the various approaches and terminologies that exist for conducting design science research. Therefore, this paper proposes a tailor-made approach for inexperienced researchers to conduct design science research. Accordingly, this approach aims to specifically assist researchers to develop an artefact as a research contribution in an environment where the artefact is developed and evaluated concurrently in close collaboration with stakeholders.

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Keywords: Design Science Research; Design-based Research; Design-oriented; Information Systems Research; Nelson Mandela University – Design Science Methodology.

1. Introduction

Humans have the ability to create and design artefacts[1] that enhance and simplify everyday life. This creation and design is evident in the present age, for example, technology is predominantly human-made [13]. It can therefore be said that designing artefacts play a major role in advancing humankind to overcome the challenges of the current age.

David Perkins [11] invited people to entertain the possibility of suddenly doing away with all human-made artefacts. The author continued to state that if this were possible, the consequences would be significant. For instance, the disappearance of clothes, roads and pavements, books and no artificial light. More so, if language itself may be considered to be design, it would also vanish [11]. It is evident from the foregoing that design is found in various fields and environments, including the information systems (IS) space.

According to Nelson and Stolterman [7], design is recognised as one of the first traditions among many traditions of humankind, including religion, science arts, and more specifically technology [14]. Technology is prominent in the IS space, and design plays a major role in the advancement of this space. Research in IS goes back more than sixty

[1] Artefacts, in the context of this paper, are typically research outputs in the form of constructs, techniques and methods, frameworks and models [15]

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years [2] and it may be said that the design aspect is one of the primary purposes of IS [1]. For this reason, many researchers believe that the design process contributes to the nature of IS [1].

Accordingly, in order to fully grasp the concept of design, it is essential to understand the design science[2] research paradigm. As such, this paper will commence with a discussion of the design science research paradigm, followed by a discussion of relevant approaches to design science research, where after the design-oriented IS research approach and the design-based research approach are combined into the design science research methodology. Finally, the application of the design science research methodology to a real-world problem, as an example, is demonstrated.

2. Design Science Research Paradigm

The term *'paradigm'* predominantly refers to an established system of views or a belief system on a particular concept [14, 4]. In the context of design science, a paradigm would suggest the existence of an established system of views or a belief system that can guide a researcher to conduct typical design science research. The primary concept of design science lies in the term *design*.

As previously mentioned, the activity of designing artefacts has been around since the beginning of time. Therefore, the process of design was previously an essential contributor to the advancement of various fields or professions, such as architecture, business, education, law and medicine [13]. However, the dependence on design for the advancement of these fields or professions regressed owing to the dominance of natural science during the last century. As such, natural science has largely removed the dependence on the design process from these fields. However, the design process still remains fundamental to the fields of management science, computer science, and chemical engineering, amongst others [13].

Nevertheless, it is important to highlight the difference between natural science and design science (also referred to as the science of the artificial)

2.1. Natural Science vs Science of the Artificial

Natural science is best described as *"a body of knowledge about some class of things, objects or phenomena, in the world: about the characteristics and properties that they have; about how they behave and interact with each other"* [13]. It is clear that the phenomena occurs naturally, or without human intervention as such.

In contrast, a phenomenon that occurs as a result of human intervention, or that is man-made, is typically termed *science of the artificial*, or in other words, *design science* [13]. Design science is best described as *"a body of knowledge about the design of artificial (man-made) objects and phenomena, artefacts, designed to meet certain desired goals"* [15]. Furthermore, design science produces knowledge typically by creating constructs, techniques and methods, frameworks and models and through the know-how of designing artefacts that will satisfy specific desired goals [15]. With this in mind, it can be said that design science research creates knowledge by using design, analysis, reflection and abstraction.

In essence, design science research may be summarised as follows: *"knowing through building"* [9]. A typical example of this would be aeronautical engineering. The knowledge base for aeronautical engineering was almost exclusively built on knowledge obtained from developing the Montgolfier balloon during WWI [15].

With the preceding discussion in mind, it is clear that design science research focuses on creating knowledge by designing artefacts. The question at this stage is, what is an appropriate approach, or methodology, to follow in order to conduct acceptable design science research within the IS space?

[2] It should be noted that literature refers to different terminologies, and sometimes these terminologies are used interchangeably [15]. Typically, the two major terms are *design research*, and *design science research*. It should be noted that there is a subtle difference between these terminologies, in that design research is research on design as a principle [15]. Design science research, on the other hand, is research whereby an output is designed through research. Therefore, for the purposes of this paper, only the term *design science research* will be used.

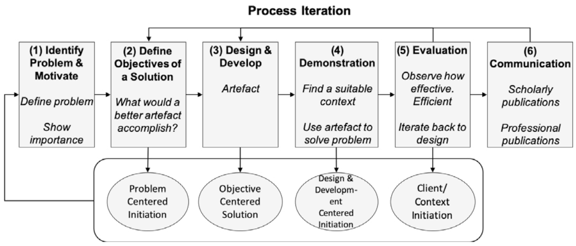

Fig. 1. Peffers' Approach - Six Activities. Adopted from Peffers et al. [10].

3. Approaches to Design Science Research

To date, various methodologies or approaches exist in terms of how design science research can be conducted [14]. The majority of these approaches have a similar goal in mind, that of designing an artefact. It should be noted that no approach or methodology supersedes another approach or methodology. It is, however, important for the researcher to choose an appropriate approach or methodology that best fits the intended research environment. As such, it is important to briefly discuss three of the more prominent approaches or methodologies (Peffers' approach, the design-oriented IS research approach and the design-based research approach) that can be used to conduct typical design science research.

3.1. The Peffers Approach

The first approach is most likely one of the better-known approaches to conducting design science research. This well-known research approach was developed by Peffers, Tuunanen, Rothenberger, and Chatterjee [10] and is referred to as the *"design science research methodology"*. Many newcomers to research confuse Peffers' approach with design science research as a paradigm. In contrast, it is important to note that design science research as a paradigm dictates research through the designing of an artefact, as described by Gregor and Hevner [5]. In order to design the required artefact, a methodology or approach is followed, which in this case is referred to as the *design science research methodology* [10], hereafter referred to as the *Peffers approach*.

The Peffers approach consists of six steps by means of which design science research is conducted. These steps will typically be followed sequentially for a problem centric research study [10]. However, a researcher is allowed to start at any step and move forward from there. Peffers et al. [10] list the six steps as follows: Step 1 – identify the problem, Step 2 – define the objectives, Step 3 – design the artefact, Step 4 – demonstrate the artefact, Step 5 – evaluate the artefact, and Step 6 – communicate the artefact. Fig. 1 clearly shows the order of Peffers' six steps.

Although the Peffers approach is well-known, it is essential to take note of Step 5 – evaluate the artefact. Once Step 1 to 4 have been completed, Step 5 requires that an evaluation of the artefact be done. This evaluation determines how well the artefact addresses the identified problem [10]. It should be noted that the evaluation typically takes place only after the artefact has been developed, as one can only then measure how well the artefact addresses the problem. This highlights the main difference between the three approaches (Peffers' approach, the design-oriented IS research approach and the design-based research approach) mentioned in this paper. The Peffers approach would therefore be more applicable for use in research that requires evaluation only after developing the artefact. Using the Peffers approach, whereby the aforementioned six steps are applied, a researcher's research can be classified as design science research.

Another prominent approach to conducting design science research is called the design-oriented IS research approach.

3.2. Design-oriented IS Research

Design-oriented IS research provides an alternative perspective on conducting design science research and essentially aims to develop and provide an artefact as a research contribution or output [8]. This artefact is very similar to that developed using the Peffers approach. Such an artefact should also aim to address a problem that exists in

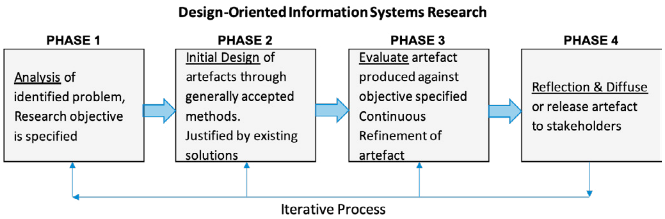

Fig. 2. Design-oriented IS Research Phases. Adapted from Österle et al. [8].

a real-world[3] environment. Furthermore, it should be noted that typically various stakeholders exist whom have an interest in the problem being addressed [8]. These stakeholders ideally participate in, and provide resources for, the research; in return, they expect favourable results for themselves [8]. Typically, these stakeholders could include economic players, such as companies and employees, public administration and all kinds of groups in society, such as students, road users, patients, and bank customers [8].

An important concept of design-oriented IS research is that the scientific rigour is substantiated by continuous refinement iterations during the artefact's creation and evaluation [8]. As such, design-oriented IS research places great importance on the stakeholders, as they form an essential part of the artefact creation and evaluation [8]. Subsequently, this approach is best suited for research where the artefact is developed and evaluated concurrently in close collaboration with stakeholders. Furthermore, design-oriented IS research advocates academic freedom and thus researchers are free to decide on objectives and research methods - this, as long as they adhere to the specific principles of design-orientated IS research [8]. These principles are briefly described as follows:

- **Abstraction:** Each artefact must be applicable to a class of problems.
- **Originality:** Each artefact must substantially contribute to the advancement of the body of knowledge. Where the body of knowledge of design-oriented IS research is constituted by the scientific literature produced and the experiences and knowledge accumulated in business [8].
- **Justification:** Each artefact must be justified in a comprehensible manner and must allow for the validation thereof.
- **Benefit:** Each artefact must yield benefits - either immediately or in the future - for the respective stakeholder groups.

The above-mentioned four principles provide the basis on which design-oriented IS research is built. This type of research, like the Peffers approach, ideally follows an iterative research process. This iterative process comprises four consecutive phases grounded on the four previously discussed principles [8]. Fig. 2 depicts these four phases.

Again, these four consecutive phases, as per Fig. 2, do not prescribe, dictate or propose comprehensive guidance to be followed and allow for *academic freedom* [8]. Thus, this academic freedom allows the researcher to select the most appropriate methods at hand. These methods should provide the researcher with detailed guidance.

Although the phases of design-oriented IS research are very similar to the steps in Peffers' approach, the main difference between Peffers' approach and design-oriented IS research is highlighted in Phase 3. In design-oriented IS research, the evaluation is a continuous process throughout the development of the artefact, as opposed to an evaluation at the end of the artefact development as proposed by the Peffers approach. Therefore, a design-oriented IS research approach would best fit a unique research environment where the artefact should be developed and evaluated concurrently with ongoing input and participation from stakeholders.

Developing this artefact might seem daunting at first, owing to design-oriented IS research providing the researcher with academic freedom. However, this approach allows the researcher to consult other methods for guidance when completing the four phases as represented in Fig. 2 [8].

[3] The term *real-world* generally refers to a problem that is deemed a realistic, addressable and current problem experienced in our modern society.

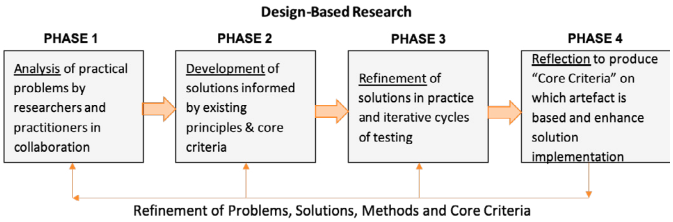

Fig. 3. Design-based Research Phases. Adapted from Reeves [12].

Accordingly, the researcher is free to consult design-based research, which is the third research approach or methodology for conducting design science research.

3.3. Design-based Research

Design-based research has a similar goal to design-oriented IS research, that of creating an artefact by considering various stakeholders' needs. The fundamental difference is that design-based research stems from the learning sciences and not IS. However, this approach includes comprehensive guidance on *how* to conduct design science research. The comprehensive guidance takes in the form of elements that have to be completed in each of the four phases.

According to Barab and Squire [3], design-based research can be defined as *"a series of approaches, with the intent of producing new theories, artefacts, and practices that account for and potentially impact learning and teaching in naturalistic settings"*. From this definition, it is clear that a very similar artefact to that of design-oriented IS research is produced. The naturalistic setting represents the research environment in which the researcher conducts research. With this in mind, design-based research also has four phases through which research is conducted [12]. Fig. 3 depicts these four phases.

Each of the four phases of design-based research include comprehensive guidance on how to complete the individual phase. As mentioned previously, the comprehensive guidance takes in the form of various elements that are contained within each individual phase. These elements act as guidance and should be addressed in order to complete the research. Herrington, McKenney, Reeves, and Oliver [6] provide a table containing each of these elements. The detail involved in applying each phase and the underlying elements of each phase will be discussed later on.

At this stage three of the more prominent approaches (Peffers' approach, design-oriented IS research and design-based research approach) have been discussed. It is clear that these approaches have similar goals in mind - that of designing an artefact. As previously stated, it is important to note that no approach or methodology supersedes another approach or methodology. It is, however, important for the researcher to choose the approach or methodology that best fits the intended research environment.

3.4. Peffers Approach vs Design-oriented IS and Design-based Research Approaches

Considering the three more prominent approaches, it is important to decide which, or probably which combination, is more suited for different research environments. As such, Tab. 1 summarises a comparison of the three approaches.

Referring to Tab. 1, it can be stated that the main difference between the three approaches is the timing of the artefact evaluation. Typically, the Peffers approach requires evaluation after the artefact has been developed (step 5). In contrast, both design-oriented IS research and design-based research approaches incorporate the evaluation as part of the development and refinement iteration process (Phase 3). As such, it is clear that the Peffers approach would best work for an environment where it is feasible to first develop the artefact after which the artefact is evaluated. In contrast, in an environment which requires the evaluation of an artefact, whilst developing the artefact in close collaboration with stakeholders, either the design-oriented IS research or the design-based research approach (or a combination of both) can be implemented. Therefore, the researcher should typically consider the research environment and implement the best suited research approach.

Table 1. Comparison of Research Approaches.

Properties	Peffers' approach	Design-oriented	Design-based
Goals	Novel artefact creation for problem-solving via invention, evaluation, measurement and impact studies. Exists in domain of IS.	Designing artefacts that is practically applicable to real-world problems. Exists in the domain of business, IS/ICT.	Novel educational technology solutions in complex situations which is generally situated in a real-world setting.
Focus	All three approaches have dual focus - that of developing artefacts and contributing to the body of knowledge.		
Distinct features	Artefacts for real-world problems. Satisficing findings. Solutions by means of human cognition.	Artefacts for real-world problems. Limited guidance, due to research methods being open for interpretation.	Rigorous inquiry into real problems in education. Clear guidance steps on process to follow.
Processes	Six-step process [10].	Four-phase process [8].	Four-phase process and guidance [6].
Application	IS and ICT.	IS and ICT.	Educational technology.
Evaluation and scientific rigour	Evaluation is done after artefact development (Step 5).	Continuous evaluation during artefact development (Phase 3).	Continuous evaluation during artefact development (Phase 3).

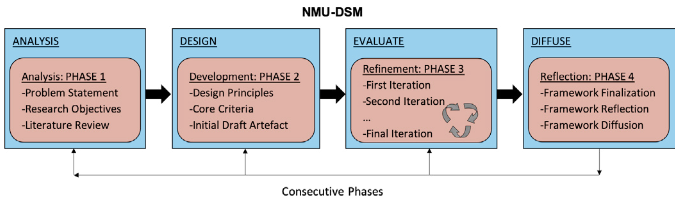

Fig. 4. Nelson Mandela University - Design Science Methodology.

With this in mind, this paper will focus on proposing a research approach for an environment which requires the evaluation of an artefact, whilst developing the artefact in close collaboration with stakeholders. Therefore, both the design-oriented IS research and the design-based research approaches should be considered.

From the preceding discussion, it is clear that both design-oriented IS research and design-based research phases have a similar goal - that of designing an artefact to address a problem in a real-world environment. The similarity of the goals may be attributed to the fact that both research approaches focus on providing an artefact as an output.

However, design-oriented IS research does not provide comprehensive guidance on how to produce the said artefact. Because of the academic freedom that design-oriented IS research provides, the researcher is free to consult design-based research, as it provides guidance in this regard. Thus, the four phases of design-based research and its underlying elements can be integrated into the four phases of design-oriented IS research, resulting in a tailor-made approach to conducting design science research in an environment where the concurrent evaluation and development of the artefact are critical. This tailor-made approach to design science research is called the *Nelson Mandela University - Design Science Methodology* (NMU-DSM).

4. Nelson Mandela University - Design Science Methodology

The NMU-DSM is aimed at providing an inexperienced researcher with guidance on conducting design science research to ultimately produce an artefact. Furthermore, the NMU-DSM is well suited for an environment which requires the evaluation of an artefact, whilst developing the artefact in close collaboration with stakeholders. Fig. 4 depicts the NMU-DSM approach to design science research. As may be seen, the NMU-DSM consists of an integration of the four phases from both design-oriented IS research and design-based research.

In order to fully understand the NMU-DSM, it is important to demonstrate the application of the four phases as presented in Fig. 4. Accordingly, the following subsection will contextualise the practical application of the NMU-DSM within a hypothetical research environment.

4.1. Applying the NMU-DSM

As mentioned previously, Herrington et al. [6] provides a table containing various elements that a researcher should implement to conduct design science research. These elements are categorised in the four main phases, as depicted in Fig. 4, and will be utilised to demonstrate the implementation of the NMU-DSM. As such, consider the following hypothetical problem statement in a real-world environment and further supported by the primary objective:

"Currently, the corporate governance of ICT (CGICT) in South African local government is unsatisfactory as highlighted by the Auditor-General of South Africa. Without good CGICT, local government is unable to effectively achieve its strategic goals, due to processes depending on ICT, and therefore not adding satisfactory value."

In order to address the stated problem, the following primary objective was identified:

"The primary objective is to produce an artefact in the form of a framework towards good CGICT (termed the F-CGICT) to assist local government with the implementation thereof."

With the aforementioned in mind, consider Phase 1 from the NMU-DSM.

4.1.1. NMU-DSM Phase 1:

The goal of Phase 1 is the *"analysis of practical problems by researchers and stakeholders in collaboration"* [6]. To do so, Phase 1 requires the researcher to complete various elements as described in Tab. 2.

Table 2. NMU-DSM Phase 1. Adapted from Herrington et al [6].

Phase of NMU-DSM	The elements that need to be completed	Position in hypothetical study
PHASE 1: Analysis of practical problems by researchers and stakeholders in collaboration.	Identification of problem statement through consultation with researchers and stakeholders.	By consulting stakeholders in local government, an initial problem was identified.
	Research objectives.	Based on the problem statement, initial research objectives were identified.
	Literature review.	From the objectives, a literature review was conducted to further understand the problem identified.

First, upon initial findings from the South African Auditor-General's report, the researcher collaborated with the local government stakeholder in order to formulate the problem statement. By collaborating with local government, the researcher identified that the problem was situated within the domain of CGICT. After formulating the problem statement, unique research objectives for the study were constructed. Furthermore, Phase 1 requires the researcher to conduct a literature review. Upon the completion of the literature review, the final output of Phase 1 is first a problem statement that has been formulated, and secondly, it presents unique objectives that have been identified that will address the problem at hand. With this output, the goal of Phase 1 has been achieved.

4.1.2. NMU-DSM Phase 2:

The main goal of Phase 2 is the development of solutions, or in this case the artefact, as informed by existing design principles (as per Sec. 3.2) and core criteria to be extracted from literature [6]. Tab. 3 represents Phase 2 and its underlying elements.

With this in mind, Phase 2 first requires the researcher to study literature and related government policy documents in order to identify design principles and core criteria that are typically required in a sound CGICT framework. Lastly, Phase 2 also requires the researcher to incorporate the identified design principles and core criteria to develop a first draft version of the artefact, or the F-CGICT version 1 in this case. As a result, the output of Phase 2 is version 1 of the F-CGICT, which aims to contribute to the identified problem from Phase 1. Subsequently, the output of version 1 of the F-CGICT concludes Phase 2.

4.1.3. NMU-DSM Phase 3:

The primary goal of Phase 3 is stated as *"iterative cycles of testing and refinement of solutions in practice"* [6]. With this in mind, Phase 3, as represented by Tab. 4, requires the researcher to refine and evaluate the artefact (F-CGICT) through various iterative cycles.

Table 3. NMU-DSM Phase 2. Adapted from Herrington et al [6].

Phase of NMU-DSM	The elements that need to be completed	Position in hypothetical study
PHASE 2: Development of solutions informed by design principles and supporting core criteria.	Firstly, identify and understand artefact design principles stemming from literature (refer to Sec. 3.2). Secondly, further identify supporting core criteria from literature to guide the design of the intervention.	Incorporate the design science principles as the foundation of the research. Then, study CGICT-related legislation, policy documents and best practices and standards to identify and extract supporting core criteria required to develop the artefact.
	Description of proposed intervention (artefact).	Develop initial draft artefact (F-CGICT version 1) from core criteria identified from principles of good CGICT.

Table 4. NMU-DSM Phase 3. Adapted from Herrington et al [6].

Phase of NMU-DSM	The elements that need to be completed	Position in hypothetical study
PHASE 3: Iterative cycles of testing and refinement of solutions in practice (continuous evaluation of artefact).	Initial development of intervention (artefact) **(First iteration)**.	First iteration starts with initial artefact (F-CGICT) as drafted in previous phase.
	Participants/stakeholders.	Members from local government (e.g. executive management).
	Data collection and data analysis.	Artefact (F-CGICT) is evaluated for acceptance. If applicable, analyse previous feedback on artefact and refine.
	Implementation of intervention.	Present draft artefact to stakeholders, refine accordingly for next iteration.
	Second and further iterations Same elements as first iteration.	Second iteration starts with version 2 of the artefact (F-CGICT) from previous iteration. Refinement and evaluation of artefact continues until acceptable level is reached, as determined by stakeholders.

The F-CGICT is refined until it has reached an acceptable level, which is determined by the stakeholders of local government. Considering the first element of Phase 3, the researcher is required to complete a first iteration of the refinement process. Subsequently, version 1 of the F-CGICT from Phase 2 is taken and presented to the stakeholders of local government. After presenting the F-CGICT, feedback is gathered. The feedback is then incorporated into a second version of the F-CGICT. After incorporating the feedback, the second iteration starts, which follows the exact pattern of the first iteration. The iterative cycles of refinement and evaluation will continue until the F-CGICT reaches an acceptable level, as determined by the stakeholders of local government [8]. The final refined F-CGICT is considered the output of Phase 3. This final refinement of the F-CGICT also concludes Phase 3.

4.1.4. NMU-DSM Phase 4:

The main goal of Phase 4 is to *"reflect on core criteria of produced artefact and enhanced solution implementation"* [6]. Essentially, Phase 4, as represented by Tab. 5, contains three elements that have to be completed.

Table 5. NMU-DSM Phase 4. Adapted from Herrington et al [6].

Phase of NMU-DSM	The elements that need to be completed	Position in hypothetical study
PHASE 4: Reflection on core criteria of produced artefact and enhanced solution implementation.	Core criteria and design principles.	Ensure F-CGICT complies with the principles and core criteria identified in Phase 2. Also validate acceptance of final artefact.
	Designed artefact(s).	Consider acceptance of final artefact and finalise the artefact, the F-CGICT in this case.
	Professional development.	Make available (diffuse) to local government as far as possible and publish solution.

Firstly, it is necessary to compare the refined F-CGICT from Phase 3 with the identified principles and core criteria from Phase 2. This comparison is done in order to check for compliance with these principles and core criteria. Secondly, if the F-CGICT complies with these principles and core criteria, the F-CGICT is finalised. Lastly, after finalisation, the F-CGICT is published. This should assist local government with the implementation of good CGICT. The output of Phase 4 is the distribution or diffusion of the final F-CGICT to the local government stakeholders. Furthermore, as mentioned previously, it is important that after the four phases of NMU-DSM have been completed, the researcher substantiate whether the F-CGICT conforms to the principles of design-oriented IS research (abstraction, originality, justification and benefit).

By following the NMU-DSM, a researcher is guided to successfully conduct design science research in an environment which requires concurrent artefact creation and evaluation in close collaboration with stakeholders.

5. Conclusion

Design science research is particularly well-known in the broader IS space. Accordingly, many researchers conduct design science research in order to develop artefacts that addresses a problem in a real-world environment. Although various methodologies or approaches exist in which design science research can be conducted, the researcher should choose an approach that best fits the intended research environment.

Unfortunately, an inexperienced researcher might find design science research a daunting task at first, due to different methodologies or approaches that exist. As such, this paper highlighted a subtle difference between three of the more prominent approaches to design science research. This subtle difference mainly determines when the artefact is evaluated. As a result, the research environment will determine which approach a researcher will use. With this in mind, this paper proposed a research approach that can guide an inexperienced researcher to conduct design science research in an environment where the evaluation of an artefact is done, whilst developing the artefact in close collaboration with stakeholders. This research approach is called the Nelson Mandela University - Design Science Methodology (NMU-DSM).

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